Social Network Analysis and Mining https://doi.org/10.1007/s13278-025-01555-3

Article in Press

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Received: 2 September 2025

Accepted: 6 November 2025

Published online: 21 December 2025

Cite this article as: Al-Gaphari G., AL-Hagree S., Al-Sanabani M. *et al.* Enhancing Arabic sentiment analysis during the Qatar world cup: a study of generative pre-trained transformers and machine learning techniques. *Soc. Netw. Anal. Min.* (2025). https://doi.org/10.1007/s13278-025-01555-3

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Enhancing Arabic Sentiment Analysis during the Qatar World Cup: A Study of Generative Pre-Trained Transformers and Machine Learning Techniques

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Abstract

Sentiment analysis plays a crucial role in understanding public opinion across Arabic social media platforms; however, challenges such as morphological richness, dialectal diversity, and contextual ambiguity hinder model performance. This study evaluates the impact of annotation quality and model architecture on the accuracy of Arabic sentiment analysis by comparing traditional machine learning algorithms with transformer-based architectures. A novel dataset was developed using an active annotation strategy that combines human expertise and artificial intelligence (AI) assisted labelling through Chat Generative Pre-Trained Transformer versions 3.5 and 4. This study investigates how annotation quality affects classification performance metrics —accuracy, precision, recall, and F1-score— across various classifiers. The experimental results reveal that annotation consistency exerts a stronger influence on performance than model complexity. Classical machine learning models such as k nearest neighbours, multinomial naïve Bayes and random forest performed best with ChatGPT 4-labelled data, reflecting the lexical coherence of AI generated annotations. Conversely, context-sensitive models including support vector machine and logistic regression, achieved superior results with human-labelled datasets, emphasizing the value of human semantic interpretation. Transformer-based architectures—AraBERTv2, CAMeL Lab BERT, and XLM-RoBERTa—outperformed all other models, demonstrating robust contextual understanding and resilience to annotation noise. The findings indicate that integrating AI-assisted annotation with transformer-based deep learning provides a scalable and linguistically adaptive framework for Arabic sentiment analysis. This study advances both theoretical understanding and practical applications by proposing a balanced human–artificial intelligence annotation paradigm that enhances the efficiency and interpretability of Arabic sentiment analysis models.

Keywords: Open artificial intelligence, Arabic sentiment analysis, Implementation of Generative Pre-Trained Transformer, Machine learning techniques, Social media data analysis, Natural language processing Applications.

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1. Introduction

The consequences of the Fédération International de Football Association (FIFA) World Cup held in Qatar in 2022 catalysed a substantial surge in Arabic online platforms debates, unveiling a distinctive opportunity for large-scale sentiment analysis (SA). However, the intricacies of Arabic language rules, morphology, and ambiguity pose substantial obstacles in analysing these conversations. These complexities have hindered the development of robust sentiment analysis (SA) models customized for Arabic text. This research sheds light on the importance of understanding public opinion in Arabic-speaking regions through enhanced sentiment analysis (SA) tools tailored to the nuances of the Arabic language. Despite these hurdles, Arabic sentiment analysis (ASA) remains a vital tool for grasping public opinion in Arabic-speaking regions.
Given the increasing digital discourse in the Arab world, accurate sentiment detection has direct implications for social research, media analytics, and policy-making, making this topic highly significant from both societal and technological perspectives.

Existing literature underscores the obstacles faced in Arabic sentiment analysis (ASA), including domain specificity, data imbalance, informal language subtleties, interpretability issues, and a reliance on traditional methods. While machine learning (ML) algorithms have shown competence in sentiment analysis, particularly in well-resourced languages such as English, Arabic unveils distinctive hurdles that demand innovative approaches for precise sentiment interpretation [1]. The complexity of Arabic grammar and morphology underscores the obstacles of sentiment analysis in Arabic text.

By addressing these challenges and leveraging advanced language models such as the Chat Generative Pre-Trained Transformer (ChatGPT) and its variants, along with other OpenAI tools, Arabic-specific pretrained models like AraBERT and CAMeLBERT, and the multilingual model XLM-RoBERTa, significant advancements can be achieved [2]. Recent studies, however, remain limited in scale and often neglect cross-model comparative analysis, creating a methodological gap that this study aims to fill.

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This research seeks to bridge existing gaps in Arabic sentiment analysis (ASA) and to enhance sentiment analysis capacity for Arabic online platforms content [3]. This research endeavours to handle the urgent demand for more refined and successful sentiment analysis tools tailored to Arabic language structures [31]. The research purposes to overcome these impediments by exploring the aptitude of ChatGPT and its new versions, sophisticated language models developed by OpenAI, to enhance Arabic sentiment analyses (ASA). Exploiting Chat Generative Pre-Trained Transformer version 4 (ChatGPT's) competence to understand and generate human-like text, the objective is to enhance sentiment analysis (SA) accuracy and efficiency in Arabic. Moreover, by comparing the performance of ChatGPT-enhanced models with fine-tuned Arabic-specific pretrained models such as AraBERT and CAMeLBERT, along with multilingual model XLM-RoBERTa, the study aims to reveal how generative transformer-based models can complement or outperform established Arabic-oriented architectures.

The research delves into the consequences of merging advanced artificial intelligence (AI) tools such as Chat Generative Pre-Trained Transformer version 4 (ChatGPT) variations with machine learning (ML) algorithms for analysing sentiments in Arabic text based on a systematic approach. A comparative examination is carried out between the Chat Generative Pre-Trained Transformer-enhanced model and traditional computational learning algorithms, including Decision Trees, Linear Regression, alongside Convolutional Neural Networks. The efficacy of these algorithms in identifying Arabic-language internet forums posts related to the Qatar World Cup is assessed. The Qatar World Cup sparked extensive arguments on internet platforms, with millions of individuals worldwide sharing their viewpoints and emotions, turning it into a global phenomenon of immense interest [4][6]. This research strives to leverage these discussions to enhance sentiment analysis methodologies in Arabic, contributing to a deeper understanding of public sentiment surrounding substantial global occurrences.

YouTube now functions as a space for individuals from diverse backgrounds to express themselves freely and share ideas [7]. This solidifies its position as a central hub for conversation and self-expression. Viewers' comments produce valuable clarity into perceptions and feedback on content, assisting producers in comprehending viewer reactions and enhancing the overall experience [8]. Marketing firms can glean insights into client expectations and sentiments by tracking these comments. Moreover, engaging in conversation enables the broader audience to learn from one another and comprehend the various perspectives shared about the video. The behaviour of the algorithms was evaluated using established machine learning (ML) metrics. A comparison of the three models revealed that machine learning (ML) algorithms augmented with Chat Generative Pre-Trained Transformer version 4 (ChatGPT) achieved better performance than the others. This research contributes remarkably to the natural language processing (NLP) domain. Initially, it develops a fresh dataset containing user comments associated with the World Cup, customized for the analysis of sentiments on YouTube. Subsequently, a comparative study is conducted involving ChatGPT version 4, ChatGPT version 3.5, and conventional sentiment analysis techniques using the same labelled dataset.

Furthermore, the inclusion of fine-tuned Arabic-specific pretrained models such as AraBERT and CAMeLBERT, together with the multilingual model XLM-RoBERTa, enriches the experimental framework and establishes comprehensive benchmarks for Arabic sentiment analysis performance. In addition, this study explores the capacity of ChatGPT version 4 and ChatGPT version 3.5 to enhance labelling precision and address the inherent challenges of analysing Arabic text. Finally, an innovative evaluation investigates the zero-shot learning capabilities of various large language models (LLMs) in performing sentiment analysis tasks, with particular emphasis on Arabic sentiment analysis.

The primary objective of this research is to evaluate the capability of ChatGPT-based generative models in enhancing the accuracy, efficiency, and interpretability of Arabic sentiment analysis, in comparison with existing Arabic-specific and multilingual pretrained architectures. Specifically, the study aims to:

1) Assess the effectiveness of ChatGPT models in classifying Arabic sentiments under both zero-shot and few-shot scenarios relative to established pretrained models.
2) Identify the strengths and limitations of generative transformer models in performing sentiment analysis across Arabic contexts, highlighting their potential to reduce dependency on extensive labeled datasets.

The paper is structured as follows: Section 2 offers a literature review on sentiment analysis focusing on the World Cup, with a focus on undertaking sentiment analysis on YouTube. Section 3 outlines the methodology employed in this study. Section 4 presents the findings and analysis of the proposed approach, including labelling by humans, ChatGPT version 3.5, ChatGPT version 4, and various combinations of these for sentiment analysis. Finally, Section 5 concludes the paper by summarizing key findings and proposing potential directions for future research.

2. Literature review

This section aims to provide a comprehensive overview of existing research on user sentiment analysis and engagement on YouTube, focusing on Arabic language content. Specifically, it explores the use of ChatGPT and its new variations to advance sentiment analysis in Arabic. The study will examine various methodologies used for sentiment analysis in YouTube videos. It will identify key challenges in Arabic sentiment analysis and determine ChatGPT's capability to address these challenges. Classifying the sentiment polarity of YouTube videos into positive, negative, or neutral, which is a primary challenge in

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sentiment analysis. YouTube's informal language and emoticons further complicate this task. This study was motivated by a desire to investigate methods ChatGPT could advance sentiment analysis in Arabic. In [1], researchers attempted to analyse the perceptions of Twitter users in Arabic-speaking countries (Egypt, Oman, Syria, Palestine, Algeria, Kuwait, Iraq, Sudan, Saudi Arabia, Jordan, Bahrain, Qatar, Yemen, and the United Arab Emirates) regarding the Qatar World Cup 2022 using a machine learning model. A redesigned dataset was constructed for this purpose. Four algorithms were employed: Random Forest (RF), Naive Bayes (NB), Logistic Regression (LR), and Support Vector Machine (SVM). LR achieved the highest average accuracy of 93%, followed by Random Forest at 92%, Naive Bayes at 88%, and SVM at 93%. In [2], various machine learning models, in conjunction with the Term Frequency-Inverse Document Frequency (TF-IDF) method, were applied to comments gathered from diverse YouTube videos about FIFA World Cup 2022. This analysis yielded an accuracy of 81.81% using Random Forest and 83.03% using Logistic Regression. This analysis produced 81.81% accuracy with Random Forest and 83.03% with LR. In [4], deep learning models such as Distilled Bidirectional Encoder Representations from Transformers (DistilBERT) and Transformer-XL Network (XLNet) were utilized to analyse Twitter sentiments about the first day of the tournament. These models were fine-tuned on a comprehensive dataset of 30,000 pre-processed messages. Model performance was evaluated using metrics like accuracy, F1-score, precision, and recall. Additionally, the study employed an Explainable Artificial Intelligence (AI) technique to understand how the model makes sentiment classification decisions. The research findings emphasize the effectiveness of these deep learning models in sentiment classification and highlight the importance of interpretability. This study contributes to our understanding of sentiment analysis during major sports events and suggests promising future research directions. In [5], sentiment analysis of messages related to the Qatar 2022 World Cup was carried out utilizing a DistilBERT model, a variant of BERT. The analysis leveraged Google Cloud Platform tools including Vertex Artificial Intelligence (AI), Storage, and BigQuery. Data collection was facilitated using the Twitter application programs interface (API), and the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology was followed. The analysis indicated a normal distribution of variables with minimal outliers. To advance future research, it's recommended to exclude outliers and fine-tune certain model hyperparameters. In [6], Textblob, a python library was used to analyses global Twitter users' sentiment regarding Qatar hosting the 2022 World Cup. The analysis was divided into three stages: before Qatar was selected as the host, after selection, and during the actual event. The findings revealed an 84% positive sentiment and a 16% negative sentiment. The model achieved an accuracy score of 0.87 (87%), indicating its capability to accurately predict 87% of the test data. Notably, the model can predict existing data without prior labelling. In [7], the authors utilized TF-IDF weighting and K-Nearest Neighbours (K-NN) to analyses Twitter sentiment surrounding the Qatar World Cup 2022. Their study provides a comprehensive overview of public opinion, including positive and negative sentiments, as well as temporal trends. The analysis revealed a positive sentiment, reflecting excitement and anticipation. However, negative sentiments also emerged, primarily related to concerns about human rights, labour practices, and the environmental impact of the World Cup. By combining TF-IDF and K-NN, this research contributes to the advancement of sentiment techniques and offers valuable insights into understanding public perception of major sporting events. A machine learning sentiment analysis solution was evaluated using a novel dataset of over 130,000 messages related to the FIFA World Cup in [8]. Using hashtags and keywords related to the Qatar World Cup, the Vader algorithm was used to analyse tweet sentiment. During the pre-tournament period, the analysis revealed a predominantly positive sentiment towards the upcoming World Cup. Researchers proposed a new method for analysing sentiment in football fan discussions in [9]. Their solution was to create a football-specific sentiment lexicon (SL) by using the specialized sentiment dataset. Finally, they trained a sentiment classifier to identify sentiments in football conversations. Through extensive experiments, they found that their approach effectively recognized fan sentiments during football events. The authors of [10] proposed a new sentiment analysis (SA) model using Global Vectors for Word Representation (GloVe) word embeddings, a specialized sentiment lexicon (SL), Multinomial Naïve Bayes Classifier (MNB), and Distilled Bidirectional Encoder Representations from Transformers (DistilBERT). The model combined Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks to capture both short-term and long-term sentiment dependencies. Experiments on the 2018 Fédération Internationale de Football Association (FIFA) World Cup messages dataset showed that the DistilBERT-CNN-LSTM model achieved high accuracy (86.5% on validation and 92.56% on testing). Additionally, the Random Forest algorithm consistently outperformed other machine learning (ML) classifiers in recognizing fan sentiments. According to [11], Twitter data was analysed using SA, relevancy analysis, and topic detection during specific events. From a corpus of over 100,000 messages, a sample of 680 was analysed. The analysis showed that 62.2% of the messages were positive, 21.8% negative, and 16.0% neutral. Additionally, 89.7% of the messages were pertinent to the events, while 10.3% were not. The study also identified similarities and differences in the topics discussed. Researchers in [12] utilized manual annotations to classify Arabic YouTube comments. They evaluated the performance of k-Nearest Neighbours (KNN), Bernoulli Naive Bayes, and Support Vector Machines with a Radial Basis Function kernel (SVM-RBF) on both raw and language-normalized datasets. The analysis revealed that SVM-RBF achieved the highest F-measure (88.8%) on the normalized dataset with two polarity classes. A four-stage process was followed in [31]: data preparation, feature extraction, optimal feature selection, and classification. During the preparation stage, techniques such as tokenization, stemming, and stop word removal were employed. To advance feature extraction beyond conventional bag-of-words methods, the researchers developed two novel features: Proposed Term Frequency-Inverse Document Frequency (Proposed TD-IDF) and advanced word2vec-based features. Additionally, a new optimization technique was introduced for optimal feature selection, aimed at improving the selection process and enhancing overall classification performance. Researchers presented a novel method for analysing Arabic

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educational videos on YouTube by using both classical ML classifiers and deep learning models [32]. Six experiments were carried out using two types of datasets: imbalanced and balanced. Three balancing techniques—oversampling, under sampling, and Synthetic Minority Oversampling Technique (SMOTE)—were applied. The analysis indicates that the Support Vector Classifier (SVC), RF, and deep learning models achieved the highest accuracy of 96% when using oversampling and SMOTE. The K-Nearest Neighbours (KNN) classifier recorded the lowest accuracy among the machine learning classifiers, at 74% with under sampling. For the deep learning model, the lowest accuracy was 89%, attained with the manually balanced dataset using bigram, trigram, and five-gram features. Additionally, a new dataset comprising Arabic YouTube educational videos was introduced, which promises to benefit researchers in the domain of data mining and related areas. For sentiment analysis in [33], three techniques were used: SentiStrength, TextBlob, and Naive Bayes Classifier. The analysis indicated that the majority of comments about the videos expressed positive sentiments, suggesting that students are satisfied with YouTube educational content and view the platform as a valuable learning tool. In [34] the study examined the number of videos related to WordPress, Joomla, and Drupal on YouTube, analysing 371, 234, and 313 videos, respectively. The outcomes indicate a steady increase in the number of videos for all three content management systems (CMSs) over four years. Currently, WordPress has the highest number of videos, followed by Drupal and Joomla. In terms of video popularity, WordPress also leads with the highest number of likes, followed by Drupal and Joomla. A sentiment analysis of comments revealed 123,409 for WordPress, 1,790 for Joomla, and 1,783 for Drupal. The analysis showed that WordPress received the highest average number of positive comments, followed by Drupal and Joomla. Additionally, in the analysis of word frequency, the word "thanks" appeared most frequently, indicating that viewers expressed a desire for more tutorial videos. In [35], the paper categorized comments from movie trailers on YouTube and established three Sentiment Indexes based on these comments to evaluate movie trailers sentiment. They used sentiment indexes to predict movie box office collections. This strategy assists distributors and filmmakers in assessing the expected response to the movie beforehand by analysing comments on the trailers. Additionally, it facilitates earnings projections on release day by factoring in current sentiments. The paper presents an innovative method of sentiment analysis and sentiment index development to forecast movie earnings. In [36], the authors evaluated the overall quality of a product. Despite this aim, the majority of comments were neutral. This study encompassed a sentiment analysis strategy involving data preparation, processing, and assessment. The model derived from this process underwent testing and assessment based on metrics such as accuracy, precision, recall, and F1-measure values. Sentiment Analysis was performed using Decision Tree and Random Forest algorithms. The Decision Tree algorithm exhibited a slightly higher accuracy rate of 89.4% than the Random Forest algorithm, which achieved 88.2%. In [37], the author provided an overview of text sentiment analysis techniques, specifically emphasizing their utilization to comments extracted from Pakistani news videos. The researchers applied a range of ML algorithms previously discussed in the paper to this dataset. The findings highlight the effectiveness of transfer learning in enhancing sentiment analysis outcomes. In [38], the authors aimed to create a comprehensive framework and practical tool for evaluating user sentiment in YouTube videos. Their method involves extracting and analysing text comments and transcripts using advanced natural language processing algorithms, classifying them into neutral, positive, negative, pertinent, or irrelevant categories. The ultimate goal is to innovate YouTube video analysis, enhancing decision-making processes and improving the overall user experience on the platform. In [39], the authors investigated the challenges of sentiment analysis (SA) on informal Arabic text, commonly found on online media platforms. While most recent sentiment analysis research focuses on English text, studies on Arabic sentiment analysis often emphasize formal Arabic. However, a significant portion of users on platforms like Twitter and YouTube express their opinions and emotions in simple Arabic (colloquial). This study examines the difficulties and challenges associated with identifying sentiment in informal Arabic, particularly as expressed in Arabic content shared on Twitter and YouTube. In [40], the authors introduced a high-quality instruction-tuning dataset, utilizing GPT-4 for efficient data generation and training of a multi-modal large language model. In [41] the authors introduced ChartLama which is a high-quality instruction-tuning dataset generated using GPT-4. This dataset was used to train a multi-modal large language model that outperformed previous methods in assessment benchmarks. In [42], the authors investigated the utilization of large language models (LLMs) to mobile software test scripts. Their goal was to address the challenges of accurately reproducing test scripts across various devices, platforms. This would advance software testing practices and quality. In [43] the authors introduced the Debiasing-Diversifying Decoding (D3) method to advance both accuracy and diversity in large language models. D3 achieves this by disabling length normalization for ghost tokens and incorporating a text-free assistant model. The comparative overview of the relevant literature is presented in Table 1.

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Table 1: comparative summary of the related literature.

Ref		Year	Dataset	Total comments or tweets	Learning	Technique
[1]		2023	Qatar World Cup 2022	17702	ML	LR, RF, NB, and SVM
[2]		2023	YouTube Comments on FIFA World Cup 2022		ML	LR, RF
[4]		2023	FIFA World Cup 2022	30,000	DL	roBERTa, distilBERT, and XLNet
[5]		2023	Qatar 2022		DL	
[6]		2024	Twitter users regarding Qatar hosting the 2022 World Cup		DL	
[7]		2023	Qatar World Cup 2022 on Twitter		ML	K-NN
[8]		2023	hashtags and keywords related to the Qatar World Cup 2022	130,000	ML	
[9]		2018	Football-Specific Tweets	54,526 tweets	ML	SVM, RF and MNB
[10]		2022	2018 FIFA World Cup		DL and ML	CNN and LSTM
[11]		2020	hashtags during specific events	100,000	ML	
[12]		2017	gathered and manually annotated YouTube Arabic comments		ML	

The examined studies on Arabic YouTube sentiment analysis, while valuable, face limitations such as domain specificity, data imbalance, challenges with informal language, lack of interpretability, and reliance on traditional techniques. To address these issues, ChatGPT can be employed for task such as adaptation, data augmentation, handling informal language understanding, developing explainable artificial intelligence models, and designing novel sentiment analysis (SA) methods. The research community is encouraged to explore these avenues by developing General-purpose models, utilizing controlled sentiment data augmentation, creating informal Arabic sentiment lexicons, integrating explainable artificial intelligence techniques, and investigating novel strategies using ChatGPT. By doing so, the domain can significantly advance SA for Arabic YouTube content, leading to a deeper understanding of user engagement and sentiment.

In this context, the present research introduces a newly developed dataset of Arabic YouTube comments related to the 2022 FIFA World Cup, along with a comprehensive multi-strategy labelling framework that applies seven distinct strategies: human annotation, ChatGPT-4, ChatGPT-3.5, and a hybrid human–ChatGPT method. The effectiveness of these labelling strategies is further evaluated using traditional machine learning algorithms, different ChatGPT versions, and fine-tuned transformer-based models, including the Arabic-specific AraBERT and CAMeLBERT, as well as the multilingual XLM-RoBERTa.

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3.Approach

This study investigates the influence of ChatGPT version 4 and ChatGPT version 3.5 on Arabic Sentiment Analysis (ASA) using a dataset of 11,373 observations. The proposed labelling procedure applies seven distinct strategies: human annotation, ChatGPT 3.5 labelling, ChatGPT 4 labelling, hybrid ChatGPT 4–ChatGPT 3.5 labelling, human–ChatGPT 3.5 labelling, and human–ChatGPT 4 labelling. These strategies are detailed in the following sections.

The overall approach consists of six major phases, as illustrated in Figure 1. A decoder-only architecture was selected for its effectiveness in sequence processing and its adaptability to open-ended generative tasks such as sentiment analysis. Supervised training was adopted due to the availability of a sufficiently large labelled dataset, ensuring optimal model performance.

3.1. Data Organization

This represents the first phase of the proposed approach. Out of the 22,000 YouTube comments initially extracted, only 11,373 aggregated Arabic comments were retained because they contained meaningful and contextually relevant content. Figure 2 illustrates the overall scraping and sentiment analysis (SA) workflow, while Algorithm 1 outlines the step-by-step process of the proposed YouTube comment downloader designed specifically for the 2022 Qatar World Cup.

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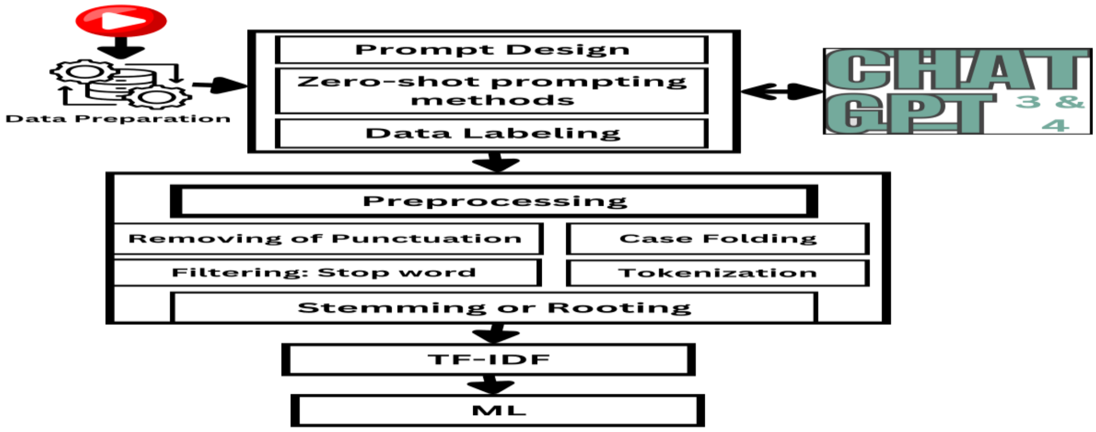

Fig. 1. The proposed methodology for data Labelling using ChatGPT version 4 and ChatGPT version 3.5 for Arabic sentiment analysis.

Data for this study were collected through a comprehensive content analysis of public YouTube videos related to the Qatar World Cup [11], [29]– [31]. YouTube was selected due to its extensive global reach and diverse user base, offering a broad range of user-generated content associated with the tournament. The dataset was using a custom Python script, ensuring careful control over the collected comments. The platform's video-centric nature allows users to express emotions, opinions, and attitudes more dynamically compared to text-only platforms, thereby providing a richer context for sentiment analysis [41], [44].
Videos were selected based on high levels of audience engagement, focusing particularly on those related to the tournament's opening ceremony and the official World Cup song [42]. These videos were identified using keyword-based searches and engagement metrics (e.g., comment count, likes, and view numbers) to ensure relevance to the study's topic. Prior research has demonstrated the efficacy of YouTube data in sentiment analysis and other social research domains [43]. Detailed procedures for ethical compliance, anonymization, and data handling are described in Section 3.2. Thus, YouTube's interactive environment was leveraged to capture public sentiment and emotional responses to the Qatar World Cup, offering a comprehensive dataset that reflects diverse Arabic-speaking perspectives [7], [8], [45], [46].

3.2 Ethical Considerations

Adhering to ethical norms is essential for maintaining the integrity, transparency, and accountability of research involving user-generated online data. This study was conducted in full compliance with established research ethics principles, particularly those concerning data sourcing, participant privacy, and responsible data handling. The following subsections describe the procedures implemented to ensure that all data were collected, reused, and analyzed in an ethically sound and methodologically rigorous manner.

3.2.1 Data Sourcing and Privacy

Data for this research were obtained exclusively from publicly accessible YouTube comments associated with the 2022 Qatar World Cup. An automated Python script employing the youtube-comment-downloader library was used to extract comments in accordance with YouTube's Terms of Service, ensuring that only publicly accessible data were collected. This process adheres to standard academic and ethical practices for data sourcing from online platforms, ensuring transparency, replicability, and respect for platform policies. No private, restricted, or user-authenticated data were accessed during this process.

3.2.2 Anonymization and Ethical Compliance

To uphold user privacy and comply with established research ethics, all personally identifiable information (PII)—including usernames, profile identifiers, and account details—was systematically removed during data preprocessing. After anonymization, the cleaned data were organized into Excel (.xlsx) format with fields for comment text, sentiment label, and video category, forming the foundation for subsequent analysis. As a result, the final dataset contains no information that could be linked to individual users. Because the study relied solely on publicly available, anonymized data and did not involve direct engagement with human participants, formal ethical approval from an institutional review board (IRB) was deemed unnecessary accordance with standard social-media research guidelines.

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Furthermore, only comments written in Arabic were retained for analysis. This linguistic filtering ensured that the research remained focused on Arabic sentiment while simultaneously reducing the dataset size and minimizing potential platform impact. The organized dataset was securely stored on local drives and documented following FAIR principles to ensure reproducibility. Consequently, the dataset produced in this study represents an ethically sound, privacy-conscious, and methodologically robust resource for advancing Arabic sentiment analysis.

3.3. Human Annotation and Dataset Creation

To construct a reliable labeled dataset, a structured human annotation process was adopted based on clear and well-defined guidelines. Three professional annotators participated in this task; all were native Arabic speakers with expertise in computer science, Natural Language Processing (NLP), and experience in sentiment analysis. Their linguistic proficiency encompassed Modern Standard Arabic (MSA) and several dialects, particularly Gulf and North African Arabic, allowing them to accurately interpret culturally nuanced expressions.

Each comment was assigned to one of three categories:

- **Positive (1):** expressions of praise, pride, or appreciation (e.g., "شكرا قطر انتي فخر لكل العرب افتتاح تاريخي" / "Thank you Qatar, you are the pride of all Arabs, a historic opening").
- **Negative (–1):** criticism, sarcasm, disapproval, or insults (e.g., "اين الملك ذيالكم اه نسيت هو لا علاقة" / "Where is your king? Oh, I forgot—he is irrelevant").
- **Neutral (0):** objective or factual remarks without a clear emotional tone (e.g., "قطر العاصمة بغداد اكبر منها" / "Qatar's capital is smaller than Baghdad").

Special attention was given to linguistic complexities. Mixed-sentiment comments were labeled according to the dominant sentiment. Sarcastic or ironic expressions were classified based on intended meaning rather than literal wording. Negation was carefully considered to avoid misclassification (e.g., "لم يكن جميلاً" / "It was not beautiful" was labeled Negative). Ambiguous, unclear, or off-topic comments were consistently labeled Neutral.

The reliability of the annotation process was quantitatively assessed using Cohen's Kappa coefficient (κ). The dataset achieved an overall $\kappa \geq 0.7$, indicating substantial agreement beyond chance. While ambiguous or mixed-sentiment cases produced lower agreement ($\kappa \approx 0.20$–0.50), strong expressions of praise or criticism yielded high agreement ($\kappa \geq 0.70$–0.85), confirming that the guidelines were applied consistently.

Disagreements were resolved through a two-step procedure: first, by majority voting among the three annotators, and second, through collective discussion when consensus was required. This ensured that each instance received a single, agreed-upon ground truth label By combining expert annotators, rigorous annotation criteria, systematic handling of ambiguous cases, and statistical validation of intercoder reliability, this process established a robust and trustworthy foundation for subsequent sentiment analysis experiments.

3.4. Large Language Models for Sentiment Analysis

The second phase of the proposed approach employs large language models (LLMs), such as Gemini and ChatGPT. These models are trained on extensive and diverse textual corpora, enabling them to capture linguistic nuances and generate coherent, human-like responses [17]–[22]. Due to their large-scale architectures—often comprising more than 100 billion parameters—LLMs can effectively process and interpret complex natural language patterns with strong contextual understanding. Typically, interaction with such models follows a two-step process: defining the specific task to be performed and providing the corresponding textual input to guide the model's output. This general workflow is schematically illustrated in Figure 2.

Algorithm 1 :YouTube Comment Downloader for Qatar World Cup 2022.

```
video IDs
For each id in video_ids:
  Initialize a YoutubeCommentDownloader object downloader
  Construct the video_URL using the id
  Get comments from the video_url with downloader and sort by popularity
  For each comment in the first 100,000 comments:
    Append the comment to a DataFrame df_comment
    Wait for 0.2 seconds
  Save df_comment to a CSV file with the id as the filename in the output_directory_path
```

Fig. 2. Algorithm1: You tube Comment Download for Qatar World Cup 2022.

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The sentiment classification of comments related to the Qatar World Cup was carried out programmatically using the official OpenAI API through the Python client library. A dedicated Python script was developed to automatically process the dataset, read each entry from Excel, submit the text to the API via the chat completions endpoint, and store the resulting sentiment outputs. Each API call instructed the model to return a single-word sentiment label—positive, negative, or neutral—without additional explanation or text.

For the classification task, two model snapshots were employed to ensure transparency and reproducibility: gpt-4-0613 (OpenAI GPT-4, June 2023 release) and gpt-3.5-turbo-1106 (OpenAI GPT-3.5, November 2023 release). Explicitly reporting these identifiers allows the experiments to be replicated precisely in future work.

To guarantee methodological rigor and eliminate randomness, the temperature parameter was fixed at 0, ensuring deterministic and consistent outputs across repeated runs. The max_tokens parameter was set to 1 to constrain the model's response to a single sentiment label. The top_p parameter remained at its default value of 1, corresponding to standard greedy decoding without introducing sampling variability. Similarly, both the frequency_penalty and presence_penalty parameters were kept at their default values of 0, meaning no penalties were applied to repeated tokens or the introduction of new words. These parameter configurations collectively ensured concise, reproducible, and methodologically sound results.

The resulting sentiment classifications were automatically recorded in a structured tabular format and exported to Excel for statistical and comparative analysis. This automated workflow, coupled with transparent reporting of model configurations, reinforces the reproducibility and scientific validity of the experimental process.

In the context of Arabic sentiment analysis, fine-tuned Arabic-specific pretrained models such as AraBERT and CAMeLBERT, together with the multilingual model XLM-RoBERTa, have demonstrated outstanding capability in capturing the semantic richness and morphological complexity of the Arabic language. These models serve as essential benchmarks for evaluating performance in Arabic sentiment analysis research. By addressing the inherent linguistic and contextual challenges of Arabic and leveraging advanced generative models such as ChatGPT and its variants, alongside other OpenAI tools, researchers can achieve significant advancements in the accuracy, interpretability, and generalization of Arabic sentiment analysis systems [2].

3.5. Prompt Design

In the third phase, the term "prompt" refers to a set of instructions used to program a Large Language Model (LLM) with the aim of guiding and optimizing its performance and capabilities. Prompts play a crucial role in influencing the model's interactions and the quality of its outputs. Consequently, accurately determining the appropriate prompt is vital to achieving the desired results for a given task. Prompts can be classified into several types, including zero-shot, one-shot, few-shot, Role, Chain, and Chain-of-Thought (CoT) variants. To identify the most effective prompt for the SA task, an initial test was conducted as detailed in references [11], [12], [13], [14], and [15]. This study specifically emphasizes zero-shot learning, which will be discussed further in the following subsection.

3.5.1. Zero-Shot Prompting Techniques of ChatGPT version 4 and ChatGPT version 3.5

Zero-shot prompting techniques are employed by ChatGPT version 4 and ChatGPT version 3.5, utilizing natural language instructions to define tasks and expected outcomes. This strategy allows the LLMs to create a context that refines the inference space, leading to more precise outputs [16]. An example of a zero-shot prompt, highlighting input instructions and placeholders for labels, is presented in Figure 3 and Table 2. Initially, simpler instructions are used in the tests, as depi cted in the figures and tables. Both ChatGPT version 4 and ChatGPT version 3.5 have the capacity to function within LLMs, with ChatGPT version 4 utilizing the same prompt as ChatGPT version 3.5

An example of a zero-shot prompt, highlighting input instructions and placeholders for labels, is presented in Figure 3 and Table 2. Initially, simpler instructions are used in the test, as depicted in the figures and tables. Both ChatGPT 4 and ChatGPT version 3.5 have the capacity to function within LLMs, with ChatGPT version 4 utilizing the same prompt as ChatGPT version 3.5.

Instructions Prompt for Arabic Sentiment Analysis (ASA):

The following prompt was used to instruct the sentiment analysis task:
"Provide sentiment analysis for the following 20 Arabic sentences in sequence. Do not include the original text—only identify one of the sentiment polarities: Positive, Negative, or Neutral."
Input Data: Table 2 presents the input comments, with examples shown in the second column.

Fig.3. An example of a Zero-shot prompting technique of ChatGPT 4 and ChatGPT 3.5.

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Table 2. Examples of Comments for ASN Based on Zero-Shot Prompting Method

ID	Translated Comment (English)	ChatGPT-4 Sentiment	ChatGPT-3.5 Sentiment
10403	The best Arab Muslim ruler, and I am proud of him — Mohamed from Casablanca, Morocco	Positive	Positive
4712	Arabs in name only; they have abandoned their Arab identity — a sad reality	Negative	Negative
4569	None of the World Cup songs are successful. If Tunisians or Algerians had made one, you'd see real creativity.	Negative	Neutral

3.5.2. Data Labeling

Data labeling (DL) is a crucial process in artificial intelligence development that involves annotating raw data with meaningful labels, making it understandable for machines and enabling deeper insight into the nuances of sarcasm [17], [18],[19] and [33]. By providing context and structure, data labeling empowers AI models to learn and improve their accuracy and mitigate sarcasm. It was divided into three approaches, as follows: Figure 4 and Algorithm 2 show the step-by-step flow of the proposed data labeling for the Qatar World Cup 2022 using ChatGPT version 4 and ChatGPT version 3.5.

Algorithm 2: DL for Qatar World Cup 2022 using of ChatGPT 4 and ChatGPT 3.5

```
1. **Initialize the environment**
   - Import necessary libraries: `streamlit`, `pandas`, `openai`, `io`, `os`.
   - Retrieve the OpenAI API key from environment variables.
2. **Setup the Streamlit interface**
   - Display the title "YouTube Comments Sentiment Labeling".
   - Provide an explanation about the application.
3. **Collect inputs from the user**
   - Input the OpenAI API key (password protected).
   - Upload an Excel file containing YouTube comments.
   - Provide a text area for a prompt template with a default example.
4. **Validate inputs**
   - Check if the API key is provided.
   - Ensure a valid Excel file is uploaded.
   - Verify the prompt template is not empty.
5. **Process the uploaded file**
   - Read the Excel file into a DataFrame. If reading fails, display an error.
   - Ensure the file has at least two columns (`ID` and `Comment`).
   - Rename columns for clarity.
6. **Setup OpenAI API parameters**
   - Configure model parameters: `temperature`, `top_p`, `frequency_penalty`, and `presence_penalty`.
7. **Iterate over the DataFrame rows**
   - For each row, extract the ID and comment.
   - Construct the final prompt by appending the comment to the template.
   - Call the OpenAI API to classify the sentiment.
   - Extract the result and append it to a list of results.
   - Handle any errors during API calls.
8. **Display and save results**
   - Convert the results list into a DataFrame with columns: `ID`, `Comment`, `Sentiment`.
   - Display the DataFrame in Streamlit.
   - Provide a download button to save results as a CSV file.
9. **Complete the process**
   - Notify the user that processing is complete and results are ready.
End of Algorithm.
```

Fig.4. Algorithm 2: DL for Qatar World Cup 2022 using of ChatGPT 4 and ChatGPT 3.5.

Steps Explanation for the Code: This section explains the steps followed in the provided Python code for implementing a **YouTube Comments Sentiment Labeling Application** using **Streamlit** and the **OpenAI API**. The application enables

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users to upload an Excel file containing comments, specify a sentiment classification prompt, and retrieve sentiment labels (positive, negative, or neutral) for each comment. The process starts by setting up the Streamlit interface to collect the OpenAI API key and the input file. Afterward, the user can customize a prompt template, which guides the sentiment classification performed by OpenAI's GPT models. The app ensures inputs are valid before proceeding. When the user triggers the sentiment labeling process, the app reads the uploaded file, iterates through each comment, and appends it to the customized prompt. It sends this prompt to OpenAI's API for classification and captures the response. Errors are logged for any failed rows. The results are stored in a Pandas DataFrame and displayed interactively in the app, with an option to download the labeled data as a CSV file. This implementation demonstrates an efficient pipeline for integrating user inputs, real-time AI-based sentiment classification, and a user-friendly web interface, making it a robust solution for text analysis applications. The algorithm is available on GitHub[*].

1)Data Labeling using Humans

This approach entails human annotators manually labelling Arabic text for sentiment analysis (SA). The findings indicate that the majority of the selected comments are categorized as positive, totalling 4,086 records. Negative records account for 753, while neutral records number 6,534. In this process, annotators carefully read and analyse the text to identify the sentiment conveyed and assign it a label of positive, negative, or neutral. This approach not only serves as a benchmark for evaluating the accuracy of other labelling methods but also relies on human evaluators who are native Arabic speakers with extensive experience across various media forms, such as videos, films, TV shows, jokes, YouTube, and social media platforms. Their exposure has honed their ability to distinguish between serious comments and sarcastic remarks. Leveraging evaluators familiarity with diverse media content enables them to effectively identify contextual cues, emotional nuances, and linguistic markers that signal sarcasm within YouTube comments. To gain deeper insight into the distribution of polarity scores within the training dataset. Figure 5 depicts the frequency distribution of polarity scores in the training dataset. Figure 6 presents the word clouds for Arabic sentiment analysis. A word cloud, or tag cloud, is a type of data visualization for text data that shows the most frequently used terms and their intensities. The larger the term, the more frequently it is used; smaller words have lower frequencies. The most common word in the dataset was Qatar (قطر). Another widely used term was Arabs (عرب). Other frequently occurring words included Ali (علي), Greetings (تحه), Salem (سلم), Allah (الله), Cup (كاس), World (عالم), Hamad (حمد), and Iraq (عرق) [25].

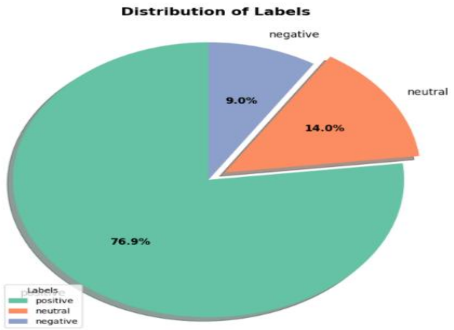

Fig. 5. The polarity score distribution for data labeling performed by humans.

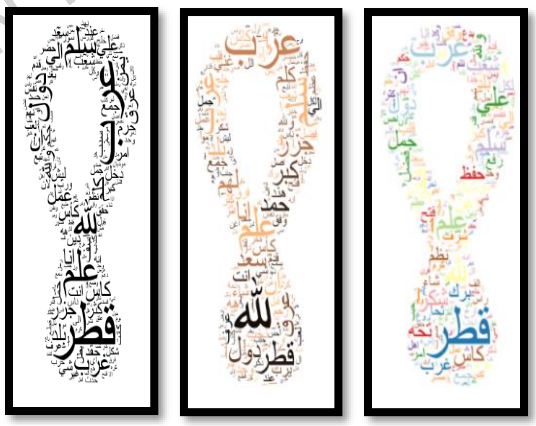

Fig. 6. Word cloud for use in comments that are positive (Right) or negative Centre) and neutral (left) for data labelling by humans.

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2)Data Labeling using ChatGPT Version 4

A novel approach to labelling for Arabic sentiment analysis involves utilizing ChatGPT version 4. This method leverages ChatGPT version 4 to label Arabic text for SA. ChatGPT version 4 is a versatile model proficient not only in Arabic NLP and in SA, but also in English NLP and various other languages. By utilizing the capabilities of ChatGPT version 4, the model's adaptability is enhanced across diverse linguistic landscapes, thereby strengthening the applicability and robustness of the Arabic sentiment analysis methodology within the unique context of the Qatar World Cup. The use of ChatGPT version 4's multilingual capabilities elevates the potential limitations in generalizing the findings to various languages and contexts.

[*] https://github.com/ALaaAlawdi/Yutube-Comments-Labeling

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With 8748 positive records, the results indicate that the majority of the chosen comments are categorized as positive. There are 1028 records that are negative and 1597 records that are neutral. This method labels Arabic text for SA using ChatGPT version 4, a cutting-edge natural language processing (NLP) model. Text can be categorized as neutral, negative, or positive sentiment using ChatGPT version 4. This approach enhances interpretability by constraining the model's output to predefined categories that are easily understandable and actionable. Despite the complexity of large language models like GPT version 4-turbo, concerns about opacity were addressed by limiting the model's output to a single word representing the sentiment of the input text. This deliberate design choice reduces the need for additional interpretability techniques, as the model's decision becomes readily observable. The primary goal of the proposed approach is to evaluate the effectiveness of labelling methods using ChatGPT version 4 for Arabic sentiment analysis, harnessing the model's advanced natural language processing abilities to enhance both accuracy and efficiency. The temperature setting of the model was kept at its default value (1.0), which balances the randomness and determinism of the model's responses. A lower temperature would make the model's responses more deterministic, while a higher temperature would increase variability. For our classification task, the default setting provided a reasonable trade-off between response diversity and reliability. Since the task involved classifying sentiments, consistency in output was crucial. The model's performance was verified through validation steps to ensure that the chosen temperature did not adversely affect the classification accuracy. By emphasizing the implicitly and directness of the output format and detailing the settings used, the model's decision-making process was made transparent and easily interpretable. Figure 7 depicts the frequency distribution of polarity scores in the training dataset. Figure 8 depicts the word clouds for Arabic sentiment analysis.

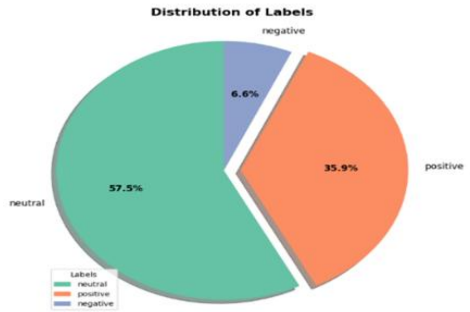

Fig.7. The polarity score distribution for data labeling performed by ChatGPT version **4.**

Fig. 8. Word cloud for use in comments that are positive (Right) or neutral (Center) and negative (left) for data labeling by ChatGPT version 4.

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3)Data Labeling using ChatGPT Version 3.5

This study introduces a novel approach to labelling Arabic sentiment analysis using ChatGPT Version 3.5. The proposed method employs this advanced natural language processing model to label Arabic text into three sentiment categories: positive, neutral, and negative. Among the labelled data, 7314 records are classified as positive, demonstrating that most selected comments fall into this category. Additionally, 1836 records are neutral, while 2223 are negative. The primary objective of this approach is to evaluate the effectiveness of ChatGPT Version 3.5 in labelling Arabic sentiment analysis. By leveraging its cutting-edge NLP capabilities, the method aims to enhance both the accuracy and efficiency of the labelling process. Figures 9 and 10 provide visual representations, with Figure 9 illustrating the polarity score distribution within the training dataset and Figure 10 showcasing word clouds for Arabic sentiment analysis.

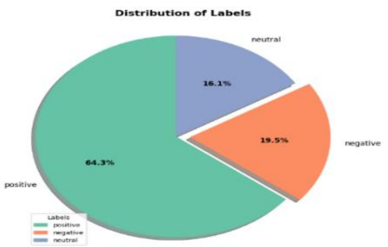

Fig. 9. The polarity score distribution for data labeling performed by ChatGPT version 3.

Fig. 10. Word cloud for use in comments that are positive (Right) or Neutral (Centre) and negative (left) for data labelling by ChatGPT 3.5

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4) Common Labeling Between ChatGPT Version 4 and ChatGPT Version 3.5

This approach involves utilizing the combined ChatGPT model (ChatGPT Version 4 and ChatGPT Version 3.5) to label Arabic text for SA. With 6739 positive records, the results indicate that the majority of the chosen comments are categorized as positive. While there are 3368 neutral records, there are 1266 negative recordings. This method entails labeling Arabic text for SA using ChatGPT, the most advanced natural language processing model. Text can be categorized as having a good, negative, or neutral sentiment using the ChatGPT version, which combines ChatGPT Version 4 and ChatGPT Version 3.5. Figure 11 depicts the frequency distribution of polarity scores in the training dataset. Figure 12 depicts the word clouds for Arabic sentiment analysis.

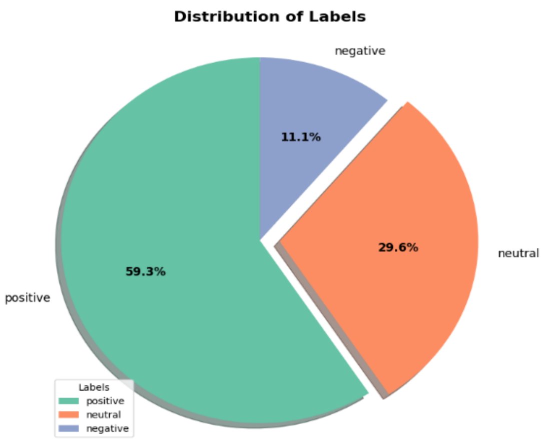

Fig. 11. The polarity score distribution for data labeling performed by ChatGPT 4 with ChatGPT version 3.5.

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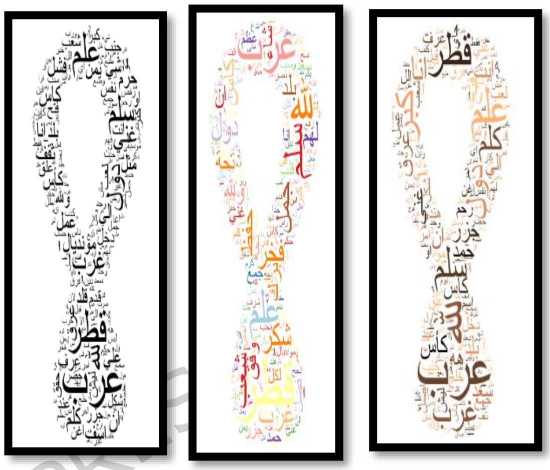

Fig. 12. Word cloud for use in comments that are positive (Right) or neutral (Centre) and negative (left) for data labelling by ChatGPT Version 4 with ChatGPT version 3.5.

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5) Common Labeling between ChatGPT Version 4 and Humans

This approach involves using the ChatGPT model (ChatGPT Version 4 and humans) to label Arabic text for SA. With 6809 positive records, the results indicate that the majority of the chosen comments are categorized as positive. While there are 2525 neutral recordings, there are 2039 negative records. This method entails labelling Arabic text for SA using ChatGPT, a cutting-edge NLP model. Text can be categorized as having a good, negative, or neutral sentiment using the ChatGPT Version 4 with people version. Figure 13 depicts the frequency distribution of polarity scores in the training dataset. Figure 14 depicts the word clouds for Arabic sentiment analysis.

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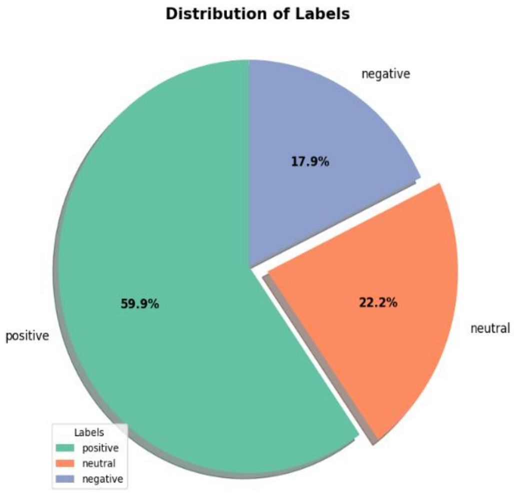
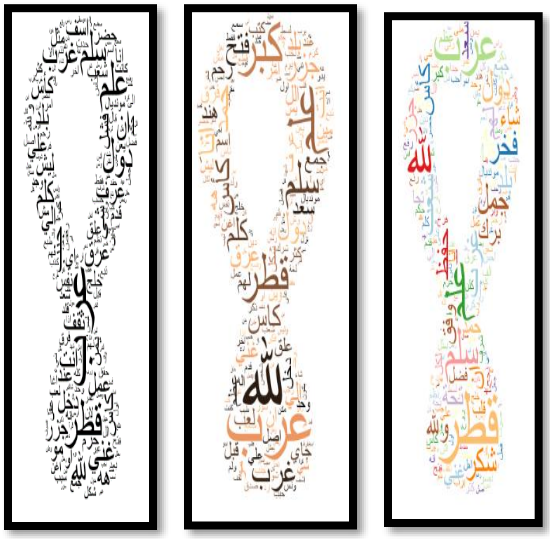


Fig. 13. The polarity score distribution for data labeling performed by ChatGPT **Version** 4 with Humans.



. **Fig. 14**. Word cloud for use in comments that are positive (Right) or neutral (Center) and negative (left) for data labeling by ChatGPT version 4 with humans.

6)Common Labeling between ChatGPT 3.5 and Humans

This approach involves using the ChatGPT model (ChatGPT Version 3.5 and humans) to label Arabic text for SA. With 7586 positive records, the results indicate that the majority of the chosen comments are categorized as positive. While there are 2485 neutral records, there are 1302 negative recordings. This method entails labeling Arabic text for SA using ChatGPT, a cutting-edge NLP model. Text can be categorized as having a good, negative, or neutral sentiment using ChatGPT Version 3.5 with human's version. Figure 15 depicts the frequency distribution of polarity scores in the training dataset. Figure 16 depicts the word clouds for Arabic sentiment analysis.

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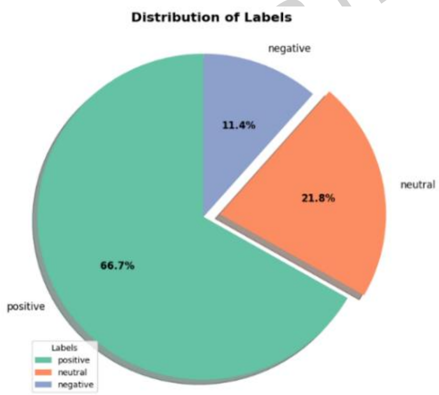

Fig. 15. The polarity score distribution for data labeling performed by ChatGPT Version 3.5 with Humans

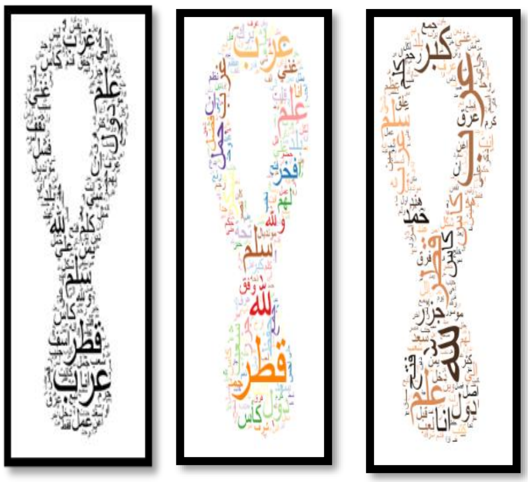

Fig.16. Word cloud for use in comments that are positive (Right) or neutral (Center) and negative (left) for data labeling by ChatGPT Version 3.5 with humans.

3.3.3. Preprocessing

The dataset is preprocessed in the fourth stage by applying several text normalization algorithms, stemming, and eliminating stop words. A critical first step in improving and deriving valuable insights from data is data pre-processing. Flaws and inconsistencies in the data, which are helped to be eliminated in this step, may affect the accuracy of the analysis.

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In the process of data pre-processing, multiple techniques play a crucial role, as outlined by references [1], [23], and [45]. An illustration
of review data is presented in Table 3, exemplifying the following procedures:

- Removal: This entails eliminating any redundant or unnecessary data that does not add to the analysis.
- Folding of Case: To cut down on the number of unique terms in the dataset, this phase entails transforming all of the text data to standard case, usually lowercase.
- Tokenization: To make analysis easier, this stage separates the text data into discrete words, or tokens.
- Stop Word Filtering: This method entails eliminating frequently used terms, also referred to as "stop words," from the dataset because they significantly diminish the analysis's usefulness.
- Rooting or stemming: By reducing the dataset's terms to their base form, this phase lowers the number of unique words and raises the analysis's accuracy.

Table 3. Example of Review data.

ID	English Translation of the Arabic Comment	Sentiment Polarity
822	This is not the World Cup opening; this is the Doha Corniche opening. Fear God.	Negative
1320	Where is the Qur’an that was used to open the World Cup?	Negative
1653	Welcome Brazil, I am from Saudi Arabia.	Positive
1835	Oh Lord, let Morocco reach the semi-finals.	Positive

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3.3.4. Features Extraction

Term Frequency Inverse Document Frequency (TF-IDF) algorithm is the fifth phase of the proposed approach. The frequency of a term in a document is measured by Term Frequency (TF), which is a key indicator for assessing a word's importance in a particular document. The number of times a term "t" occurs in the document divided by the total number of terms in the document is the TF for that term. The different document lengths are taken into consideration by this standardization. As shown in Eq (1).

$$TF(t,d) = \frac{Number\ of\ time\ t\ term\ appears\ in\ document\ d}{total\ number\ of\ terms\ in\ the\ document\ d} \quad (1)$$

Inverse Document Frequency (IDF) evaluates a term's significance within a corpus or collection of texts. It assists in determining a term's prevalence or rarity across all publications. The logarithm of the total number of documents in the corpus divided by the number of documents that contain the phrase is how the IDF for a term is calculated. Terms that appear frequently in papers are given less weight by this calculation, making them less important. As shown in Eq (2).

$$IDF(t,c) = log(\frac{total\ number\ of\ \ documents\ in\ corpus\ C}{total\ number\ of\ documents\ contining\ terms\ t})\ (2)$$

The significance of the words in the dataset is ascertained using this approach. It calculates each word's TF-IDF value according to how frequently it appears in the dataset's documents. Words that appear in fewer papers have a higher TF-IDF value, while words that appear in more documents have a lower TF-IDF value. This method lessens the influence of frequent words that do not have any unique meaning and assesses the importance of terms in describing the text dataset. [26], [27].

3.3.5. Machine Learning (ML)

In the sixth stage, machine learning (ML), a branch of artificial intelligence, entails creating algorithms that let computer systems learn from data and gradually get better at what they do without explicit programming. To identify if a text is neutral, positive, or negative, SA and ML are used to identify the sentiment expressed in the text. ML models like as LR, DT, RF, SVM, RidgeClassifier, MNB, and K-NN can all be applied to SA. Every algorithm has distinct advantages and disadvantages, and the properties of the data being analysed might affect how well an algorithm performs [24].

1) Decision Tree (DT)

A well-liked machine learning approach for classification and regression applications is the decision tree (DT). It uses a structure like a tree to model decisions and their potential outcomes. At each node of the tree in a DT, the dataset is divided

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according to several properties or features. In order to maximize the homogeneity or purity of the generated subsets, the algorithm chooses the appropriate attribute to split the data depending on specific criteria (such as information gain or Gini impurity). Recursively dividing the data, the tree keeps growing until a halting condition is satisfied. This could be when there are no more attributes to divide, a minimum number of instances per leaf, or a predetermined depth limit. Each leaf node represents a class or a predicted value for regression tasks. During the prediction phase, a new instance is traversed down the tree by following the decision paths based on the attribute values. The final prediction is determined by the class or value associated with the leaf node reached. DT have several advantages. They are easy to understand and interpret, as the resulting tree structure can be visualized. DT can handle both categorical and numerical data and are robust against noise and missing values. They can also capture non-linear relationships between features. However, DT are prone to overfitting, where they memorize the training data excessively. To overcome this, techniques like pruning or using ensemble methods such as RF can be employed [3].

2) Random Forest (RF)

RF is a popular ML model that is used for both classification and regression tasks. It is an ensemble learning method that combines multiple DT to make predictions. In a RF, a collection of DTs is created, where each tree is trained on a different subset of the data. To build each DT, a random subset of features is selected for each split, hence the term "random" in Random Forest. This random feature selection helps to introduce diversity among the trees and reduce the risk of overfitting. During the prediction phase, each DT in the RF independently makes predictions, and the final prediction is determined by majority voting (in classification) or averaging (in regression) of the individual tree predictions. RF have several advantages. They can handle high-dimensional datasets with a large number of features, and they are robust against overfitting. They are also capable of capturing complex relationships between variables and handling missing data. Additionally, RF provides estimates of feature importance, offering insights into the relative significance of different features in the prediction process.

3) Support Vector Machines (SVM)

SVM are machine learning (ML) classification techniques that accommodate for incorrect classification by transferring a space of data points that are not linearly separable onto a new space using a kernel function. For Arabic sentiment analysis, it has been used successfully [28].

4) Multinomial Naïve Bayes Classifier (MNB)

MNB classifier is a variant of NB classifier, which is employed with discrete features like word counts in text classification tasks, specifically for sentiment analysis problems. In this work, the MNB is adopted for the SA problem.

5) K-Nearest Neighbors (K-NN)

The K-Nearest Neighbours (K-NN) classification technique estimates the conditional distribution of a label variable Y given an input matrix X. The algorithm assigns a class label to an input instance based on the highest probability among the classes. The first step in the K-NN process involves calculating the distance between the input instance and the labelled reference instances in the training data. This is accomplished using a distance metric, with the Euclidean distance being one of the most commonly utilized metrics. The Euclidean distance between two points, u=(u1,u2,...,up) and v=(v1,v2,...,vp)), is defined as follows [45]:

$$d(u,v) = \sqrt{\sum_{i=1}^{p}(u_i - v_i)^2} \qquad (3)$$

K-NN employs a distance metric to identify k observations from a dataset that are nearest to an input instance x0, given X as the input matrix and Y as the label variable. The algorithm then estimates the conditional probability of x0belonging to a specific class using the following equation, where K is a positive integer:

$$Pr(Y = j|X = x_0) = \frac{\sum_{i \in N_0} I(y_i = j)}{k} \qquad (4)$$

In this case, $I(y_i=j)$ is an indicator variable that equals 1 if (x_i,y_i) is a member of class j and 0 otherwise, whereas N_0 is the set of k-nearest instances. K-NN assigns the input instance x_0 to the class with the highest probability after determining the probabilities.

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6) Logistic Regression (LR)

Based on the data, Logistic Regression (LR) calculates a class's conditional probability distribution. No prior knowledge is assumed. The classifier is forced to find the ME distribution that satisfies the constraint since the ratio of the training dataset limits the conditional distribution. Because our dataset is short and the classification is binary, LRCV is used in conjunction with a solver (liblinear) to determine the parameter weights in order to minimize a cost function [28].

4.Experiments

This section presents the experimental procedures, benchmark metrics, and performance analysis conducted in this study. It is organized into three subsections: the experimental setup (sub section 4.1), the assessment benchmark (sub section 4.2), and the analytical results (sub section 4.3). These experiments were designed to evaluate the effectiveness of both traditional and transformer-based models in Arabic sentiment analysis under various annotation strategies.

4.1 Experimental Setup

Sentiment analysis (SA) on Arabic text was conducted using two main model categories. The first category comprised traditional Machine Learning (ML) algorithms, including Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), Ridge Classifier, Multinomial Naïve Bayes (MNB), and K-Nearest Neighbors (K-NN). The second category consisted of transformer-based deep learning models, namely AraBERTv2, CAMeL-Lab BERT, and XLM-RoBERTa, fine-tuned on the same labelled datasets.

All models were trained and evaluated using six annotation strategies to examine how labelling sources influence model performance: human annotation, ChatGPT 3.5 annotation, ChatGPT 4 annotation, hybrid ChatGPT (ChatGPT 3.5 with ChatGPT 4) annotation, human with ChatGPT 3.5 annotation, and human with ChatGPT 4 annotation.

Uniform preprocessing steps were applied across all experiments, including text normalization, tokenization, and stop word removal. For transformer-based models, text sequences were tokenized using each model's pre-trained tokenizer and truncated to a fixed sequence length. All models were evaluated on identical data splits to ensure fair comparison. ML algorithm hyperparameters were tuned via grid search, while transformer-based models used fixed epochs, learning rates, and batch sizes. Cross-validation was employed to guarantee robust and unbiased evaluation across all methods.

4.2 Assessment Benchmark

The efficiency of the suggested approach, that involved human labelling, labelling by ChatGPT 4, and labelling by ChatGPT 3.5, was assessed in this research. Multiple assessment metrics, including accuracy, precision, recall, and F-score, were leveraged. Prior to the assessment of the prediction system's performance, the dataset was divided into testing and training sets, as outlined in Equations (5), (6), (7), and (8).

$$Accuracy = \frac{TP+TN}{TP+FP+TN+FN} \quad (5)$$

$$Recall = \frac{TP}{TP+TN} \quad (6)$$

$$Precision = \frac{TP}{TP+FN} \quad (7)$$

$$F-score = \frac{2*Recall*Precision}{Recall+Precision} \quad (8)$$

4.3 Machine Learning Based Analysis

As depicted in Table 4, Table 5, Table 6, Table 7, Table 8, Table 9, and Table 10, seven algorithms—K-NN, DT, LR, RF, MNB, RidgeClassifier, and SVM—were leveraged to classify the sentiment of each comment as positive, negative, or neutral. Subsequently, the trained algorithms were then applied to the testing set in order to test their performance. The optimal outcomes in terms of accuracy and precision were acquired after splitting the dataset into 80% for training and 20% for testing. The K-Nearest Neighbours (K-NN) algorithm was employed to evaluate the performance of Arabic sentiment analysis (ASA) under different labelling strategies. Table 4 summarizes the outcomes of these strategies, revealing that labelling by ChatGPT Version 4 achieved the highest overall performance among the tested approaches. Specifically, ChatGPT Version 4 achieved an accuracy of 75.00%, precision of 70.00%, recall of 75.00%, and an F-score of 71.00%. These metrics highlight the effectiveness of ChatGPT Version 4 in generating accurate and consistent labels for sentiment analysis tasks, outperforming both human labelling and hybrid labelling methodologies. Human labelling, while competitive, showed slightly lower performance compared to ChatGPT Version 4, with an accuracy of 73.00%, precision of 74.00%, recall of 73.00%, and an F-score of 70.00%. This result reflects the challenges of manually labelling datasets in a way that aligns with the K-NN algorithm's assumptions, particularly in capturing the complex sentiment nuances of the Arabic language. Other automated approaches, such as labelling by ChatGPT Version 3.5, demonstrated reasonable performance but fell short of ChatGPT Version 4. ChatGPT Version 3.5 achieved an accuracy of 70.00%, with a precision of 68.00% and an F-score of 68.00%. While these results indicate the potential of ChatGPT Version 3.5 as a labelling tool, the noticeable gap in performance compared to ChatGPT Version 4 underscores the advancements in labelling quality and consistency introduced by the latter model. Hybrid approaches, combining human labelling with AI, yielded mixed results. The combination of human labelling with ChatGPT Version 4 achieved an accuracy of 66.00%, while integration with ChatGPT Version 3.5 resulted in an accuracy of 68.00%. Similarly, combining ChatGPT Version 4 with ChatGPT Version 3.5 yielded an accuracy of 67.00%. These results suggest that hybrid methods may introduce additional variability in the dataset, potentially leading to inconsistencies that diminish performance. In deduction, those findings indicate that labelling by ChatGPT Version 4 represents an efficient strategy for enhancing Arabic sentiment analysis.

Table 4. Arabic sentiment analysis results using K-NN Technique

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Approach’s	Accuracy	Precision	Recall	F-score
labelling by Humans	0.73	0.74	0.73	0.7
labelling by ChatGPT 3.5	0.70	0.68	0.7	0.68
labelling by ChatGPT 4	**0.75**	**0.7**	**0.75**	**0.71**
labelling by ChatGPT 4 with ChatGPT 3.5	0.67	0.65	0.67	0.66
labelling by Humans with ChatGPT 3.5	0.68	0.65	0.68	0.66
labelling by Humans with ChatGPT 4	0.66	0.65	0.66	0.65

The Multinomial Naïve Bayes (MNB) technique was employed to evaluate the performance of Arabic sentiment analysis (ASA) using various labelling approaches. Table 5 presents the results, indicating that labelling by ChatGPT 4 achieved the highest performance metrics among all methods tested. Specifically, ChatGPT Version 4 achieved an accuracy of 77.00%, precision of 64.00%, recall of 77.00%, and an F-score of 67.00%. These results suggest that ChatGPT Version 4 provides a robust and effective labelling solution for sentiment analysis tasks when used with the MNB classifier. Interestingly, human labelling also produced competitive results, achieving an accuracy of 76.00%, a slightly lower precision of 71.00%, and an F-score of 73.00%. While the results for human labelling are strong, the slightly higher recall and overall accuracy of ChatGPT Version 4 indicate its ability to better generalize sentiment patterns within the dataset, making it particularly effective for this task. Hybrid approaches, combining human labelling with AI-assisted methods, showed mixed results. For instance, combining ChatGPT Version 4 with human labelling achieved an accuracy of 65.00%, which was lower than both standalone ChatGPT Version 4 and human labelling. Similarly, integrating ChatGPT Version 3.5 with human labelling resulted in an accuracy of 68.00%. These findings highlight that hybrid labelling approaches may introduce inconsistencies or noise that reduce their effectiveness unless carefully managed. ChatGPT Version 3.5 alone achieved an accuracy of 68.00% and an F-score of 59.00%, falling short compared to ChatGPT Version 4. This result demonstrates the improvements in labelling quality and consistency offered by newer AI models like ChatGPT Version 4. Furthermore, combining ChatGPT Version 4 with ChatGPT Version 3.5 produced an accuracy of 67.00%, suggesting that blending two AI models may not always lead to performance improvements and could introduce overlapping errors. The findings underscore the potential of ChatGPT Version 4 as a reliable tool for labelling Arabic sentiment datasets when paired with the MNB classifier. While human labelling remains a strong benchmark, the efficiency and scalability of ChatGPT Version 4 make it an attractive option for large-scale sentiment analysis projects.

Table 5: Arabic sentiment analysis results using the MNB technique.

Approach’s	Accuracy	Precision	Recall	F-score
labelling by Humans	0.76	0.71	0.76	0.73
labelling by ChatGPT 3.5	0.68	0.68	0.68	0.59
labelling by ChatGPT 4	**0.77**	**0.64**	**0.77**	**0.67**
labelling by ChatGPT 4 with ChatGPT 3.5	0.67	0.7	0.67	0.6
labelling by Humans with ChatGPT 3.5	0.68	0.58	0.68	0.56
labelling by Humans with ChatGPT 4	0.65	0.67	0.65	0.56

The Random Forest (RF) technique was employed to analyse Arabic sentiment using various labelling approaches. Table 6 summarizes the results, revealing that the labelling approach utilizing ChatGPT Version 4 achieved the best overall

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performance. Specifically, ChatGPT Version 4 demonstrated an accuracy of 77.00%, precision of 59.00%, recall of 77.00%, and an F-score of 67.00%. These results indicate that ChatGPT 4 effectively identifies sentiment patterns in Arabic text, outperforming both human labelling and other automated or hybrid methods in this context. Interestingly, while human-labelled datasets typically outperform automated approaches in many tasks, they yielded a lower accuracy of 61.00% in this experiment. The F-score for human labelling was 50.00%, primarily due to a mismatch between the RF model's assumptions and the characteristics of the manually labelled dataset. This suggests that while human labelling excels in nuanced sentiment analysis, the RF algorithm may be better suited to the patterns captured by AI-assisted methods, particularly ChatGPT Version 4. Hybrid labelling approaches combining humans and AI models showed mixed results. For instance, the combination of human labelling with ChatGPT Version 3.5 achieved an accuracy of 67.00%, while the integration of humans with ChatGPT Version 4 resulted in a lower accuracy of 60.00%. These findings imply that hybrid approaches may dilute the effectiveness of distinct labelling strategies unless carefully optimized. Similarly, the combination of ChatGPT Version 4 with ChatGPT Version 3.5 produced an accuracy of 59.00%, reflecting a potential conflict in the patterns learned by the two models. ChatGPT Version 3.5 alone achieved an accuracy of 64.00%, showing a moderate level of performance but falling short compared to ChatGPT Version 4. The performance disparity across these approaches highlights the potential of ChatGPT Version 4 as an effective labelling tool for Arabic sentiment analysis when paired with the RF algorithm. However, the results also underscore the need for careful consideration of the alignment between the chosen algorithm and the labelling approach.

Table 6: Arabic sentiment analysis results using RF technique.

Approach's	Accuracy	Precision	Recall	F-score
labelling by Humans	0.61	0.69	0.61	0.5
labelling by ChatGPT 3.5	0.64	0.41	0.64	0.5
labelling by ChatGPT 4	**0.77**	**0.59**	**0.77**	**0.67**
labelling by ChatGPT 4 with ChatGPT 3.5	0.59	0.35	0.59	0.44
labelling by Humans with ChatGPT 3.5	0.67	0.45	0.67	0.53
labelling by Humans with ChatGPT 4	0.6	0.36	0.6	0.45

The Support Vector Machine (SVM) technique was employed to assess the performance of Arabic sentiment analysis (ASA) across different labelling approaches. As illustrated in Table 7, human-labelled datasets consistently outperformed other methods, achieving an accuracy of 81.00%, precision of 80.00%, recall of 81.00%, and an F-score of 79.00%. These results emphasize the critical role of human expertise in achieving high-quality sentiment analysis, particularly in a linguistically complex language such as Arabic. Automated labelling approaches showed varied performance levels, with ChatGPT Version 4 achieving an accuracy of 78.00%, surpassing ChatGPT Version 3.5, which attained an accuracy of 76.00%. Despite these results, both methods demonstrated lower precision and F-scores compared to human-labelled data, reflecting challenges in accurately capturing subtle sentiment cues. Hybrid approaches, where human labelling was combined with AI-assisted methods, yielded mixed results. For example, the integration of human labelling with ChatGPT Version 4 resulted in an accuracy of 74.00%, while combining human labelling with ChatGPT Version 3.5 achieved a slightly lower accuracy of 72.00%. This indicates that while AI can complement human efforts, the integration does not necessarily lead to superior results without careful calibration. Interestingly, the combination of ChatGPT Version 4 and ChatGPT Version 3.5 produced an accuracy of 73.00%, showing that blending two AI models did not surpass the performance of individual approaches. This suggests that automated methods, even when combined, may struggle with the inherent linguistic and contextual challenges present in Arabic sentiment analysis. Overall, these findings highlight the strength of human-labelled datasets for achieving superior performance in SVM-based sentiment analysis. While AI models like ChatGPT Version 4 offer a viable alternative, particularly for scalability, they require further refinement to match the nuanced understanding provided by human annotators. Future work should explore advanced hybrid models that incorporate AI's efficiency with human expertise to enhance the accuracy, precision, and overall performance of Arabic sentiment analysis.

Table 7. Arabic sentiment analysis results using SVM technique.

Approach's	Accuracy	Precision	Recall	F-score
labelling by Humans	**0.81**	**0.8**	**0.81**	**0.79**
labelling by ChatGPT 3.5	0.76	0.74	0.76	0.74
labelling by ChatGPT 4	0.78	0.73	0.78	0.7
labelling by ChatGPT 4 with ChatGPT 3.5	0.73	0.72	0.73	0.71
labelling by Humans with ChatGPT 3.5	0.72	0.69	0.72	0.68
labelling by Humans with ChatGPT 4	0.74	0.73	0.74	0.73

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The Logistic Regression (LR) technique was employed to evaluate Arabic sentiment analysis (ASA) performance across different labelling approaches. As shown in Table 8, labelling by humans yielded the highest performance metrics, achieving an accuracy of 80.00%, precision of 80.00%, recall of 80.00%, and an F-score of 79.00%. These results underscore the importance of human expertise in capturing the intricacies of Arabic sentiment, making it the most effective method among the approaches tested. AI-assisted labelling methods demonstrated varying degrees of effectiveness. Labelling by ChatGPT Version 4 achieved an accuracy of 78.00%, slightly outperforming ChatGPT Version 3.5, which achieved an accuracy of 77.00%. While these AI-based methods offered competitive results, their inability to match human-labelled datasets highlights the nuanced understanding that human annotators bring to the task. Hybrid labelling approaches, combining human efforts with AI, provided mixed results. For instance, the combination of human labelling with ChatGPT Version 4 achieved an accuracy of 74.00%, similar to the standalone ChatGPT Version 4 approach. However, blending human labelling with ChatGPT Version 3.5 resulted in a lower accuracy of 71.00%. Similarly, the combination of ChatGPT Version 4 with ChatGPT Version 3.5 yielded an accuracy of 74.00%, indicating that merging AI models without human input may not significantly enhance performance. The findings suggest that while AI-based methods, particularly ChatGPT Version 4, are valuable for scalable and efficient labelling, they are not yet sufficient to fully replicate the quality of human annotations. The slightly lower F-scores observed in AI-based and hybrid approaches indicate challenges in maintaining consistent precision and recall across diverse sentiment expressions

Table 8: Arabic sentiment analysis results using LR technique.

Approach's	Accuracy	Precision	Recall	F-score
labelling by Humans	**0.80**	**0.8**	**0.8**	**0.79**
labelling by ChatGPT 3.5	0.77	0.75	0.77	0.74
labelling by ChatGPT 4	0.78	0.76	0.78	0.72
labelling by ChatGPT 4 with ChatGPT 3.5	0.74	0.72	0.74	0.72
labelling by Humans with ChatGPT 3.5	0.71	0.68	0.71	0.67
labelling by Humans with ChatGPT 4	0.74	0.73	0.74	0.72

The RidgeClassifier technique was employed to assess the performance of Arabic sentiment analysis (ASA) with various labelling approaches. As presented in Table 9, the results indicate that human-labelled datasets outperformed all other approaches. Specifically, labelling by humans achieved an accuracy of 80.00%, precision of 80.00%, recall of 80.00%, and an F-score of 79.00%. These findings underscore the capability of human annotators to accurately capture the subtle nuances of sentiment in the Arabic language, resulting in robust performance metrics. Among the AI-assisted labelling methods, the use of ChatGPT Version 4 showed notable performance, achieving an accuracy of 78.00% and a recall of 78.00%. However, while this approach-demonstrated improvement over ChatGPT 3.5 (accuracy of 75.00%), it still fell short compared to human labelling. Combining labelling strategies, such as integrating ChatGPT Version 3.5 or ChatGPT Version 4 with human-labelled datasets, resulted in mixed outcomes. For example, the combination of human labelling with ChatGPT Version 4 achieved an accuracy of 74.00%, which, while respectable, did not surpass the standalone human-labelled dataset. Interestingly, the combination of ChatGPT Version 4 with ChatGPT Version 3.5 yielded an accuracy of 72.00%, indicating that blending AI models alone is less effective than incorporating human expertise. Similarly, the integration of ChatGPT Version 3.5 with human labelling achieved an accuracy of 71.00%, highlighting a diminishing return when relying too heavily on automated methods. These findings emphasize the critical role of human expertise in achieving high accuracy and precision in ASA tasks. Although AI-based labelling offers scalable and efficient solutions, it struggles to match the nuanced understanding provided by human annotators. Future work could explore optimized hybrid approaches that combine the scalability of AI with the accuracy of human annotations, potentially unlocking further advancements in sentiment analysis for Arabic and other complex languages.

Table 9: Arabic sentiment analysis results using RidgeClassifie technique.

Approach's	Accuracy	Precision	Recall	F-score
labelling by Humans	**0.80**	**0.8**	**0.8**	**0.79**
labelling by ChatGPT 3.5	0.75	0.73	0.75	0.73
labelling by ChatGPT 4	0.78	0.74	0.78	0.73
labelling by ChatGPT 4 with ChatGPT 3.5	0.72	0.7	0.72	0.7
labelling by Humans with ChatGPT 3.5	0.71	0.68	0.71	0.68
labelling by Humans with ChatGPT 4	0.74	0.72	0.74	0.72

The performance of Arabic sentiment analysis was evaluated using the decision tree (DT) technique with various labelling approaches. The results, summarized in Table 10, demonstrate that the approach of labelling by humans achieved superior outcomes compared to other methods. Specifically, labelling by humans yielded an accuracy of 75.00%, precision of

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74.00%, recall of 75.00%, and an F-score of 75.00%. These metrics highlight the effectiveness of human-labelled datasets in capturing the nuances of Arabic sentiment. In contrast, automated labelling approaches, including those using ChatGPT models, produced comparatively lower performance. Labelling with ChatGPT Version 3.5 achieved an accuracy of 68.00%, while ChatGPT Version 4 showed slight improvement with an accuracy of 70.00%. Combining labelling approaches, such as integrating ChatGPT Version 3.5 with ChatGPT Version 4 or human labels, resulted in marginally lower performance, with accuracies ranging from 66.00% to 66.00%. These results suggest that while AI-assisted labelling can be helpful, it does not yet surpass the accuracy and precision of human labelling for Arabic sentiment analysis tasks. Overall, the findings emphasize the importance of human expertise in ensuring high-quality sentiment analysis in the Arabic language. While AI-based models like ChatGPT can assist in labelling, their current capabilities fall short of fully replicating the nuanced understanding that human annotators bring. Future research should explore hybrid approaches that leverage the strengths of both human expertise and AI to further enhance performance.

Table 10: Arabic sentiment analysis results using DT technique.

Approach's	Accuracy	Precision	Recall	F-score
labelling by Humans	**0.75**	**0.74**	**0.75**	**0.75**
labelling by ChatGPT 3.5	0.68	0.67	0.68	0.67
labelling by ChatGPT 4	0.70	0.7	0.7	0.7
labelling by ChatGPT 4 with ChatGPT 3.5	0.66	0.65	0.66	0.66
labelling by Humans with ChatGPT 3.5	0.66	0.64	0.66	0.65
labelling by Humans with ChatGPT 4	0.66	0.65	0.66	0.65

While both approaches exhibited decent performance, ChatGPT Version 4 generally outperformed ChatGPT Version 3.5 in the Arabic sentiment analysis task when utilizing the various ML techniques tested, including K-NN, MNB, and RF. Although the human labelling approach demonstrated better performance, it typically surpassed ChatGPT Version 4 and ChatGPT Version 3.5 in the Arabic sentiment analysis task when employing the diverse ML techniques tested, such as SVM, LR, DT, and RidgeClassifier.

4.4. transformer-based analysis

Tables 11–13 present the performance of the transformer-based deep learning models—AraBERTv2, CAMeL-Lab BERT, and XLM-RoBERTa—across the six annotation strategies. Overall, the results reveal that transformer architectures achieved notably higher accuracy and consistency than traditional machine learning algorithms, confirming their strong ability to capture complex semantic and contextual nuances in Arabic text.

1) AraBERTv2

The AraBERTv2 model was fine-tuned on six differently labelled Arabic sentiment datasets to evaluate how annotation sources influence performance. As shown in Table 11, the human-labelled dataset achieved the highest overall scores, with an accuracy, precision, recall, and F-score all equal to 0.84. This balance across evaluation metrics indicates that manual annotations provided consistent and contextually rich sentiment cues that aligned effectively with AraBERTv2's linguistic understanding. When ChatGPT Version 3.5 was used for annotation, the model maintained strong performance (accuracy of 0.81), demonstrating that GPT-generated labels can approximate the quality of human annotations. In contrast, ChatGPT Version 4 achieved a lower accuracy (0.73) but displayed an outstanding precision of 0.88, suggesting that its predictions were more confident and selective despite fewer correct classifications overall.

Hybrid labelling strategies produced mixed results. Combining ChatGPT 4 and ChatGPT 3.5 yielded moderate outcomes (accuracy of 0.77, F-score of 0.74), likely due to inconsistencies between the two annotation styles. A more promising result emerged when human annotations were merged with ChatGPT 3.5, achieving an F-score of 0.84—comparable to the fully human-labelled dataset. This synergy indicates that ChatGPT 3.5 can complement human annotations by reinforcing sentiment patterns frequently observed in social media text. Meanwhile, the human–ChatGPT 4 combination produced balanced results (accuracy and F-score of 0.80), showing solid generalization though with slightly less harmony than the 3.5-based hybrid.

Overall, these findings highlight that AraBERTv2 performs best with high-quality human annotations, yet it remains robust when trained on AI-generated or hybrid-labelled data. The exceptional precision of ChatGPT 4 and the balanced performance of the human–ChatGPT 3.5 combination demonstrate that transformer-based models can benefit from carefully integrated human–AI collaboration. This confirms AraBERTv2's adaptability for Arabic sentiment analysis, provided that annotation consistency and contextual richness are maintained across labelling sources.

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Table 11 Arabic sentiment analysis results using AraBERTv2 Model

Approach’s	Accuracy	Precision	Recall	F-score
labelling by Humans	**0.84**	**0.84**	**0.84**	**0.84**
labelling by ChatGPT 3.5	0.81	0.81	0.81	0.81
labelling by ChatGPT 4	0.73	0.88	0.78	0.83
labelling by ChatGPT 4 with ChatGPT 3.5	0.77	0.76	0.78	0.74
labelling by Humans with ChatGPT 3.5	0.76	0.90	0.79	0.84
labelling by Humans with ChatGPT 4	0.80	0.81	0.80	0.80

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2) CAMeL-Lab BERT

The CAMeL-Lab BERT model was fine-tuned on six versions of the Arabic sentiment dataset, each annotated through distinct strategies to assess the influence of labelling quality on model outcomes. As presented in Table 12, the human-labelled dataset achieved the most balanced and consistent performance across all metrics, with accuracy, precision, recall, and F-score each recording 0.82. This demonstrates that manually curated annotations provided coherent sentiment signals that aligned well with the model’s contextual and linguistic representation of Arabic text.

Annotations generated by ChatGPT Version 3.5 resulted in an accuracy of 0.76, showing a moderate reduction in performance but still maintaining strong alignment with the human-labelled benchmark. ChatGPT Version 4 produced slightly improved results in accuracy (0.77) and recall (0.77), though with a minor dip in precision (0.74), indicating that it generalized well but was somewhat less selective in classification. These outcomes suggest that while both GPT-based annotations are effective, subtle variations in label consistency and context interpretation influence the model’s predictive stability.

Hybrid labelling strategies displayed diverse results. The combination of ChatGPT 4 with ChatGPT 3.5 yielded an accuracy of 0.76 and an F-score of 0.75—values that reflect moderate performance, likely affected by overlapping annotation biases between the two models. When human annotations were combined with ChatGPT 3.5, the model’s accuracy decreased to 0.72, suggesting that divergent labelling patterns may have introduced noise in sentiment boundaries. Similarly, the human–ChatGPT 4 combination achieved a uniform score of 0.75 across all metrics, indicating improved balance but not surpassing the purely human-labelled dataset.

Overall, CAMeL-Lab BERT demonstrated high sensitivity to annotation quality, excelling with consistent human labels and maintaining commendable robustness when trained on AI-assisted data. The findings affirm that while GPT-based annotations offer scalability and efficiency, human supervision remains crucial for preserving contextual integrity in Arabic sentiment analysis. Carefully managed human–AI collaboration thus provides a promising direction for future transformer-based sentiment research.

Table 12. Arabic sentiment analysis results using CAMeL-Lab-Model

Approach’s	Accuracy	Precision	Recall	F-score
labelling by Humans	**0.82**	**0.82**	**0.82**	**0.82**
labelling by ChatGPT 3.5	0.76	0.76	0.76	0.76
labelling by ChatGPT 4	0.77	0.74	0.77	0.75
labelling by ChatGPT 4 with ChatGPT 3.5	0.76	0.75	0.76	0.75
labelling by Humans with ChatGPT 3.5	0.72	0.71	0.72	0.71
labelling by Humans with ChatGPT 4	0.75	0.75	0.75	0.75

3) XLM-RoBERTa

The XLM-RoBERTa model was fine-tuned on six variations of the Arabic sentiment dataset to assess its adaptability across different labelling strategies. As shown in Table 13, the human-labelled dataset produced the most stable and balanced outcomes, achieving 0.82 across all performance metrics—accuracy, precision, recall, and F-score. This result underscores the advantage of manually curated annotations, which provide nuanced contextual cues that align well with XLM-RoBERTa’s multilingual architecture and contextual embedding capabilities.

When ChatGPT Version 3.5 was used for annotation, the model maintained strong performance, achieving an accuracy of 0.80 and a precision of 0.81. These results suggest that ChatGPT 3.5 is capable of generating labels that approximate the semantic consistency of human annotation, enabling the model to effectively capture sentiment polarity within Arabic social media discourse. By contrast, ChatGPT Version 4 yielded slightly lower results (accuracy of 0.75 and F-score of 0.74), implying that

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while its annotations were coherent, minor inconsistencies or over-generalizations may have affected the model's predictive precision.
The hybrid labelling approaches showed moderate but dependable outcomes. The combination of ChatGPT 4 and ChatGPT 3.5 achieved balanced results (accuracy and F-score of 0.78), suggesting complementary yet distinct annotation patterns that preserved model stability. When human annotations were merged with ChatGPT 3.5, the performance declined to an accuracy of 0.72 and an F-score of 0.73, indicating potential variation between human sentiment interpretation and GPT-generated labels. In contrast, combining human annotations with ChatGPT 4 improved performance slightly to 0.77 across all metrics, reflecting a more consistent alignment between the two labelling styles.
Overall, XLM-RoBERTa demonstrated resilience across all labelling scenarios, confirming its capability to generalize effectively from both human and AI-generated annotations. While human-labelled datasets remain the gold standard for maximizing contextual fidelity, the results highlight the potential of ChatGPT-based annotation—particularly when harmonized with human oversight—to support scalable and reliable Arabic sentiment analysis using multilingual transformer models.

Table 13. Arabic sentiment analysis results using XLM-Model

Approach's	Accuracy	Precision	Recall	F-score
labelling by Humans	**0.82**	**0.82**	**0.82**	**0.82**
labelling by ChatGPT 3.5	0.80	0.81	0.81	0.80
labelling by ChatGPT 4	0.75	0.73	0.75	0.74
labelling by ChatGPT 4 with ChatGPT 3.5	0.78	0.78	0.78	0.78
labelling by Humans with ChatGPT 3.5	0.72	0.75	0.72	0.73
labelling by Humans with ChatGPT 4	0.77	0.77	0.77	0.77

4.5 Comprehensive Analysis and Interpretation

The comparative experiments between traditional ML and transformer-based approaches reveal that annotation quality exerts a stronger influence on Arabic sentiment classification than model complexity. Transformer architectures, however, display built-in mechanisms that mitigate inconsistencies arising from imperfect labelling.

4.5.1 Machine Learning-Based Insights

ML models such as K-NN, MNB, and RF performed best with ChatGPT 4-labelled data (accuracy ≈ 0.75–0.77), reflecting their reliance on lexical regularity and frequency-based features that benefit from AI's consistent annotation style. Conversely, SVM, LR, and RidgeClassifier achieved superior accuracy (≈ 0.80–0.81) with human-labelled data, as human annotators better capture irony, idioms, and contextual cues essential for semantic separation. Hybrid labelling reduced accuracy (0.65–0.74) due to noise propagation—an outcome consistent with the theory that increased label entropy weakens generalization by disturbing feature–class mappings. Hence, scalable AI-assisted annotation remains useful, but human oversight is crucial for semantically rich contexts.

4.5.2 Transformer-Based Insights

Transformer models—AraBERTv2, CAMeL-Lab BERT, and XLM-RoBERTa—outperformed all classical ML methods (F-score ≈ 0.80–0.84). Their contextual embeddings and multi-head self-attention dynamically weight semantic relationships, enabling accurate sentiment inference even with noisy labels. AraBERTv2 achieved the highest adaptability (F-score 0.84 with human data), while ChatGPT 4 labelling improved precision (0.88) at the expense of recall. Such outcomes exemplify transformers' semantic robustness under weak supervision—their capacity to redistribute semantic weight to preserve context despite imperfect annotation.

4.5.3 Theoretical Synthesis and Implications

Transformers exceeded ML models by 7–10% in accuracy due to their ability to model sentiment as continuous semantic relationships rather than discrete lexical patterns. The findings align with representation learning theory, confirming that high-dimensional contextual embeddings reduce dependence on label precision. Moreover, the results validate the emerging view that ChatGPT 4-generated annotations introduce lexical regularities beneficial to statistical learners, while human annotations remain indispensable for encoding cultural and pragmatic subtleties.
In synthesis, the integration of AI-assisted annotation and transformer-based contextual learning offers a scalable paradigm for Arabic sentiment analysis—balancing automation efficiency with linguistic fidelity and interpretive depth.

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5 Conclusion

This study provides a comprehensive investigation into Arabic sentiment analysis (ASA) by integrating traditional machine learning (ML) algorithms and transformer-based models under various annotation strategies. It advances prior research by examining how annotation quality—not merely model complexity—governs the accuracy and interpretability of sentiment classification in morphologically rich languages such as Arabic.

The experimental findings reveal that annotation consistency exerts a decisive impact on model performance. Classical ML methods, including K-NN, MNB, and RF, achieved optimal results when trained on ChatGPT 4-labelled data, benefiting from lexical uniformity and reduced intra-class variance of AI-generated annotations. Conversely, context-dependent classifiers such as SVM, LR, and RidgeClassifier performed best with human-labelled datasets, highlighting the importance of human semantic discernment in capturing cultural and pragmatic nuances often lost in automated labelling.

Transformer-based architectures—specifically AraBERTv2, CAMeL-Lab BERT, and XLM-RoBERTa—demonstrated superior adaptability and semantic robustness. Their contextual embeddings and attention mechanisms allowed them to sustain high performance even under noisy or partially automated annotations, achieving F-scores between 0.80 and 0.84. This resilience reflects their ability to internalize sentiment polarity through distributed semantic relationships rather than explicit label precision.

The comparative analysis establishes a hierarchical relationship between annotation quality, model architecture, and contextual understanding. It confirms that while AI-assisted annotation using ChatGPT 4 offers scalability and lexical coherence, human supervision remains essential for preserving interpretive depth and cultural context. The integration of both human and AI annotation strategies, when coupled with transformer-based deep learning, represents a balanced and scalable framework for advancing ASA.

From a theoretical perspective, these results reinforce the robustness principle of contextual embeddings and contribute to the emerging discourse on LLM-driven annotation consistency. Practically, they highlight the potential of ChatGPT 4 as an effective annotation assistant for large-scale sentiment datasets, enabling cost-efficient yet reliable labelling for under-resourced languages.

Limitations and Future Work:

Despite its contributions, this study has certain limitations. The dataset is limited to text-based Arabic tweets and does not account for multimodal cues such as emojis, tone, or images that often influence sentiment interpretation. Additionally, while the annotation comparison between human and ChatGPT 4 is insightful, it does not include other large models like GPT-4o or Gemini, which may yield different behaviors. Future work should extend the analysis to larger and more diverse corpora, incorporate multimodal sentiment features, and assess domain adaptation across dialectal Arabic and cross-platform datasets. Integrating explainability tools such as SHAP or LIME can also enhance model transparency and interpretability in real-world applications.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT for checking statements syntax and words spelling in order to improve English writing. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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